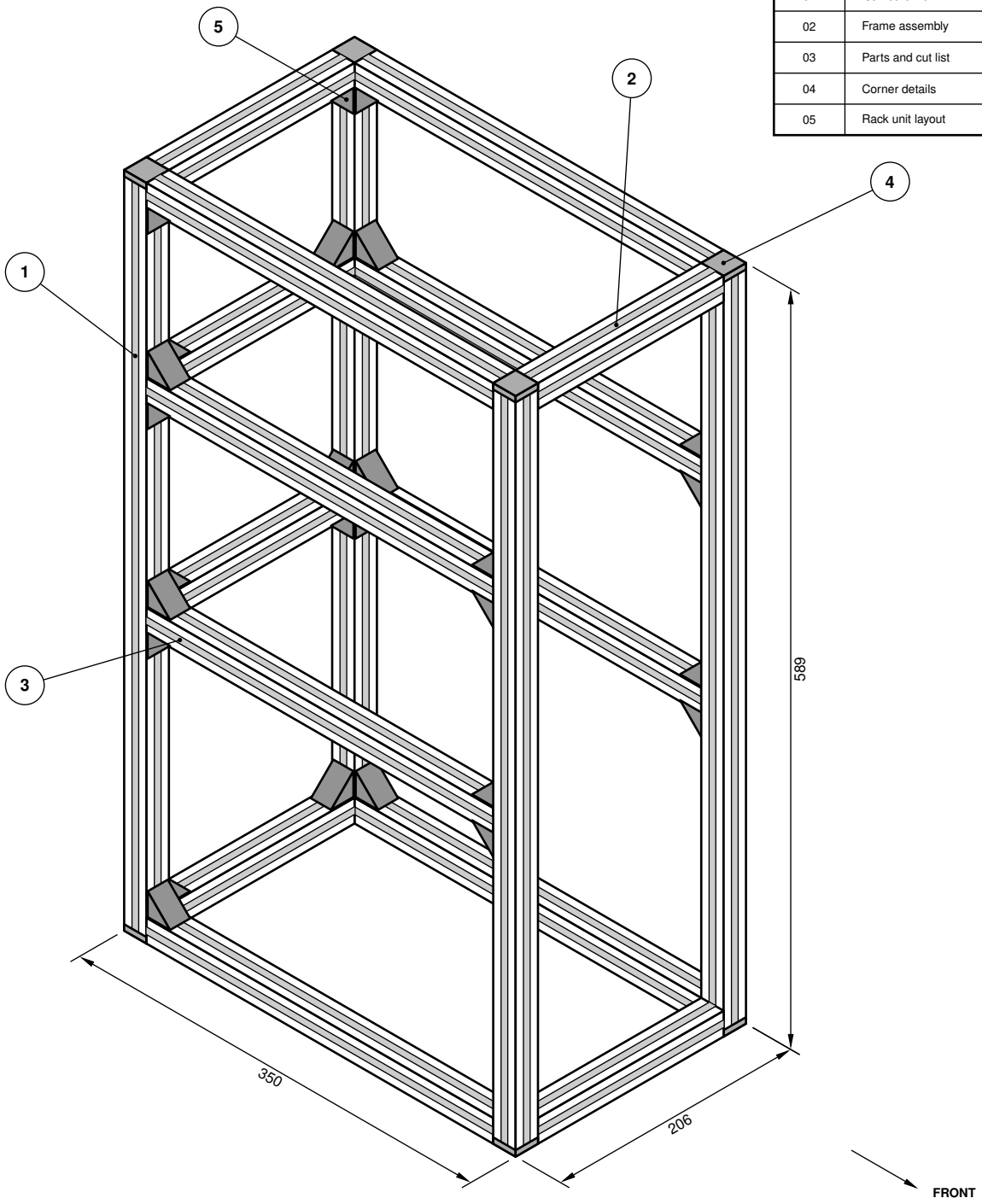
## ISOMETRIC VIEW 1:5

| SHEET | TITLE |
|---|---|
| 01 | Isometric view |
| 02 | Frame assembly |
| 03 | Parts and cut list |
| 04 | Corner details |
| 05 | Rack unit layout |




### NOTES

1. All dimensions are in mm; keep in mind that 1U = 1.75 in = 44.45 mm.
2. Isometric drawing at 1:5 on all three axes. Hidden edges are omitted. Slots shaded in light grey, corner connectors in dark grey, L brackets in near black.
3. The overall height of 589 includes the 5 mm corner connectors at the top and bottom vertices; uprights are cut to 579.
4. Item numbers match sheets 02 to 04. Orthographic views on sheet 02, cut lengths on sheet 03, corner details on sheet 04, rack unit layout on sheet 05.
5. Intermediate depth rails and back width rails at the U3/U4 and U7/U8 boundaries.
6. L brackets (5) on every upright to rail joint of the sides and the back, none at the four front vertices: 32 in all. Drawn schematically.

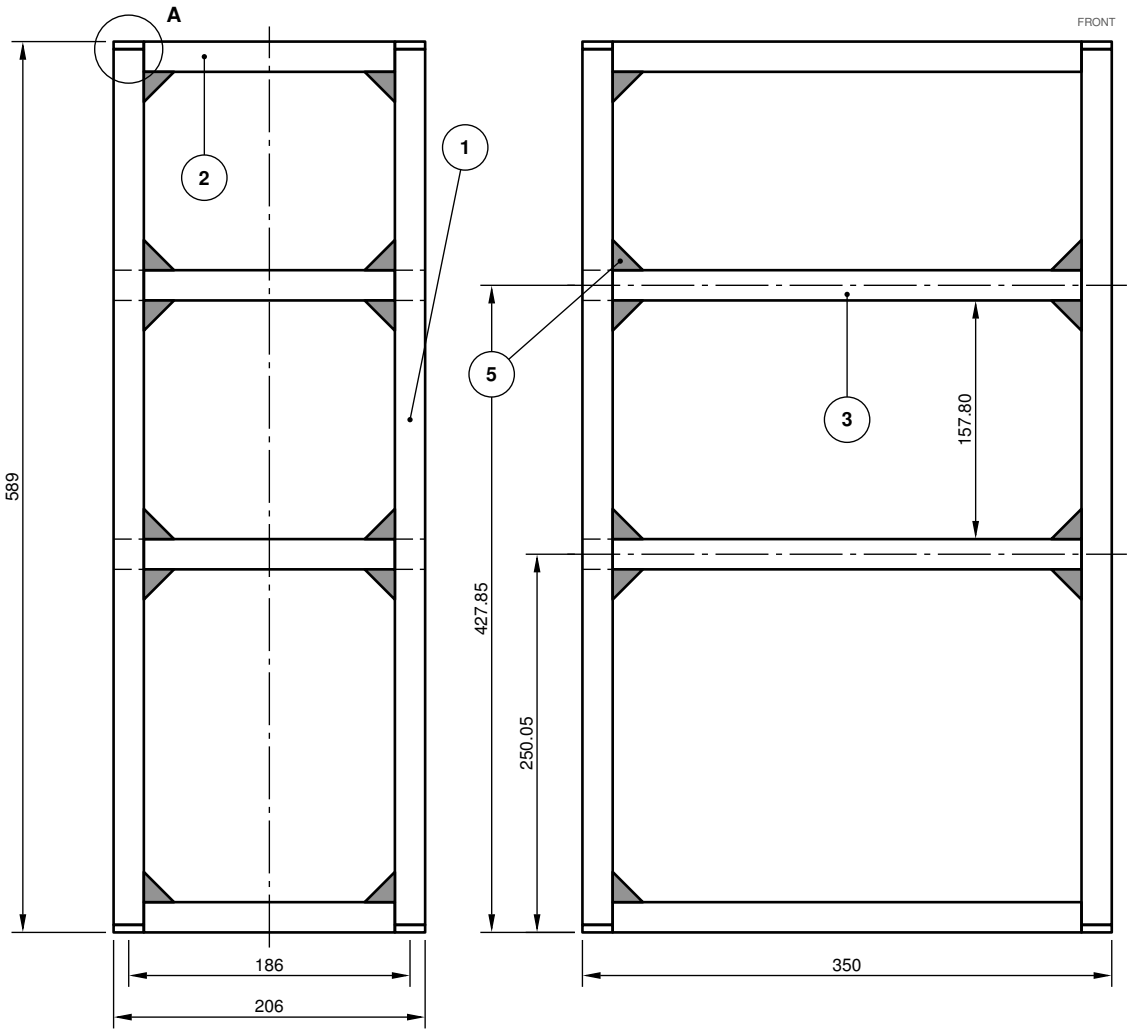
OWNER

**LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION**

| TITLE | PROJECTION | UNIT |
|---|---|---|
| **ISOMETRIC VIEW**<br>Frame overview, 8 in. × 12U format | **ISOMETRIC** | **mm** |

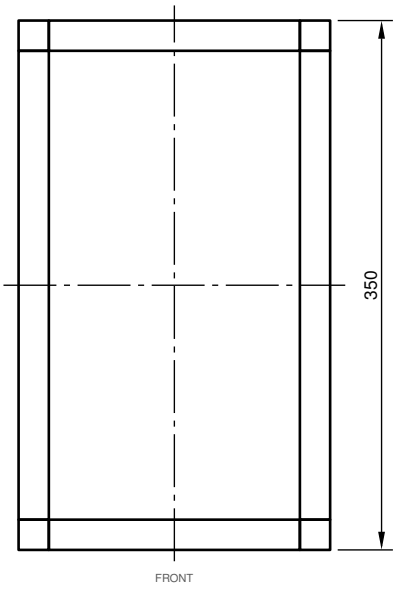
| MATERIAL | STANDARDS |
|---|---|
| Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot | ISO 5457 · 7200 · 128 · 129-1 |

| DRAWING NO. | SCALE | SHEET | DATE | REV. |
|---|---|---|---|---|
| **MR2020-01** | **1:5** | **1 / 5** | 2026-09-13 | **A** |

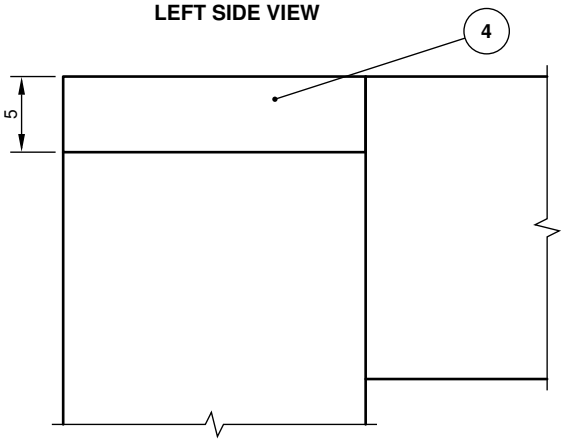



**FRONT VIEW**

**LEFT SIDE VIEW**




**TOP VIEW**




**DETAIL A (2:1)**

5 mm connector at each upright end, bottom mirrored

**NOTES**

1. All dimensions are in mm; keep in mind that 1U = 1.75 in = 44.45 mm.
2. Heights measured from the frame base, at the bottom face of the lower connectors.
3. Each corner connector adds 5 mm, at the top and at the bottom vertices: uprights are cut to 579 and the frame totals 589. See detail A.
4. First angle projection. The front is on the right in the side view and at the bottom in the top view.
5. Intermediate depth rails and back width rails at the U3/U4 and U7/U8 boundaries.
6. Cut lengths on sheet 03.
7. L brackets (5) at every upright to rail joint of the sides and the back, none at the front vertices, which stay clear for the rack panels. Each takes two hammer nuts (6) and two screws (7).

| ITEM | DESCRIPTION | QTY | NOTE |
|---|---|---|---|
| 1 | Upright | 4 | cut 579 |
| 2 | Width rail | 6 | cut 166 |
| 3 | Depth rail | 8 | cut 310 |
| 4 | 3-way corner connector | 8 | 4 in stock |
| 5 | L corner bracket, 2 holes | 32 | 16 in stock |
| 6 | Hammer nut, 20 series | 64 | 8 in stock |
| 7 | Button head hex screw, 8 mm | 64 | purchased |

OWNER

**LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION**

TITLE

**FRAME ASSEMBLY**

20 × 20 extrusion, 8 in. × 12U format

PROJECTION

UNIT

**mm**

MATERIAL

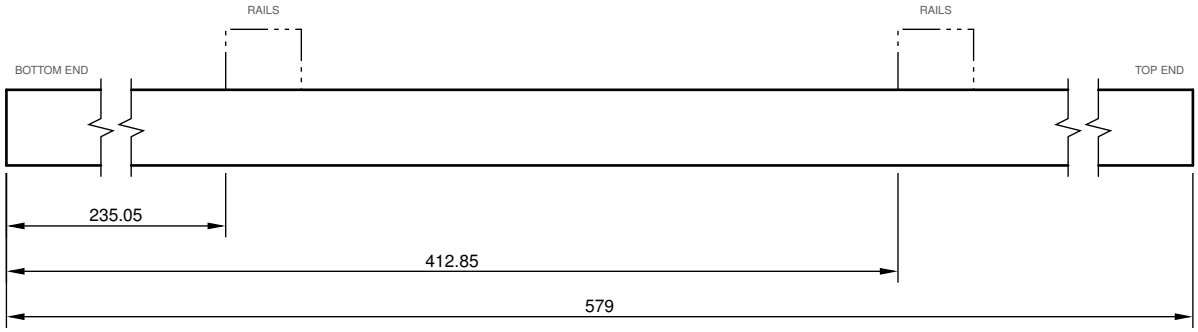
Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot

STANDARDS

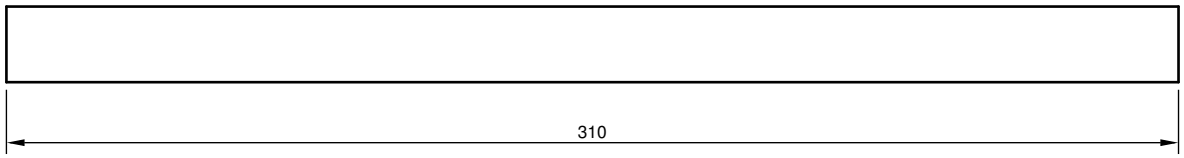
ISO 5457 · 7200 · 128 · 129-1

| DRAWING NO. | SCALE | SHEET | DATE | REV. |
|---|---|---|---|---|
| **MR2020-02** | **1:5** | **2 / 5** | 2026-09-13 | **A** |

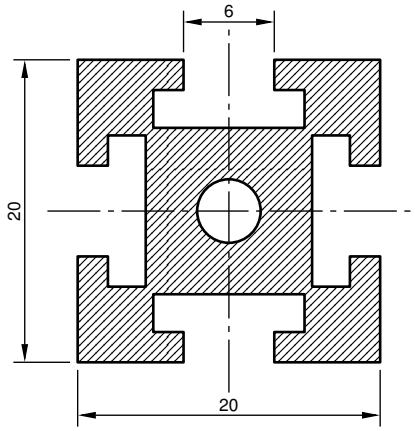
**ITEM 1 UPRIGHT QTY 4** **1:2**



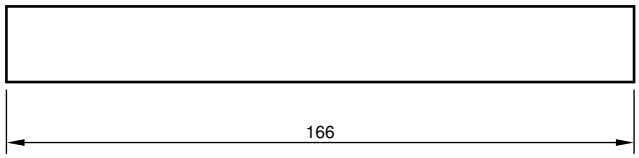

**ITEM 3 DEPTH RAIL QTY 8** **1:2**




**PROFILE SECTION** **2:1**




**ITEM 2 WIDTH RAIL QTY 6** **1:2**




**CUT LIST**

| ITEM | DESCRIPTION | QTY | LENGTH | TOTAL |
|---|---|---|---|---|
| 1 | Upright | 4 | 579 | 2316 |
| 2 | Width rail | 6 | 166 | 996 |
| 3 | Depth rail | 8 | 310 | 2480 |
| | | | **TOTAL PROFILE** | **5792** |

**NOTES**

1. All dimensions are in mm. General length tolerance ±0.5.
2. Cut square: the corner connectors depend on it. Deburr both ends of every piece.
3. Lengths do not include saw kerf. Measure before each cut.
4. Upright positions are measured from the bottom end of the upright to the lower face of each rail. Sheet 02 gives the rail axes from the frame base: subtract 5 mm for the lower connector and 10 mm for half a rail.
5. The corner connectors add 5 mm at each end of the upright and are not part of the cut length.
6. On the back uprights the same marks also locate the back intermediate width rails.
7. The upright is drawn as an interrupted view; the dimensions give the true lengths.
8. Profile section is schematic. Slot and bore geometry follow the supplier catalog.
9. Purchased hardware (corner connectors, L brackets, hammer nuts and screws) is listed on sheet 02.

OWNER

**LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION**

TITLE

**PARTS AND CUT LIST**

Cut pieces for the 8 in. × 12U frame

PROJECTION

UNIT

**mm**

MATERIAL

Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot

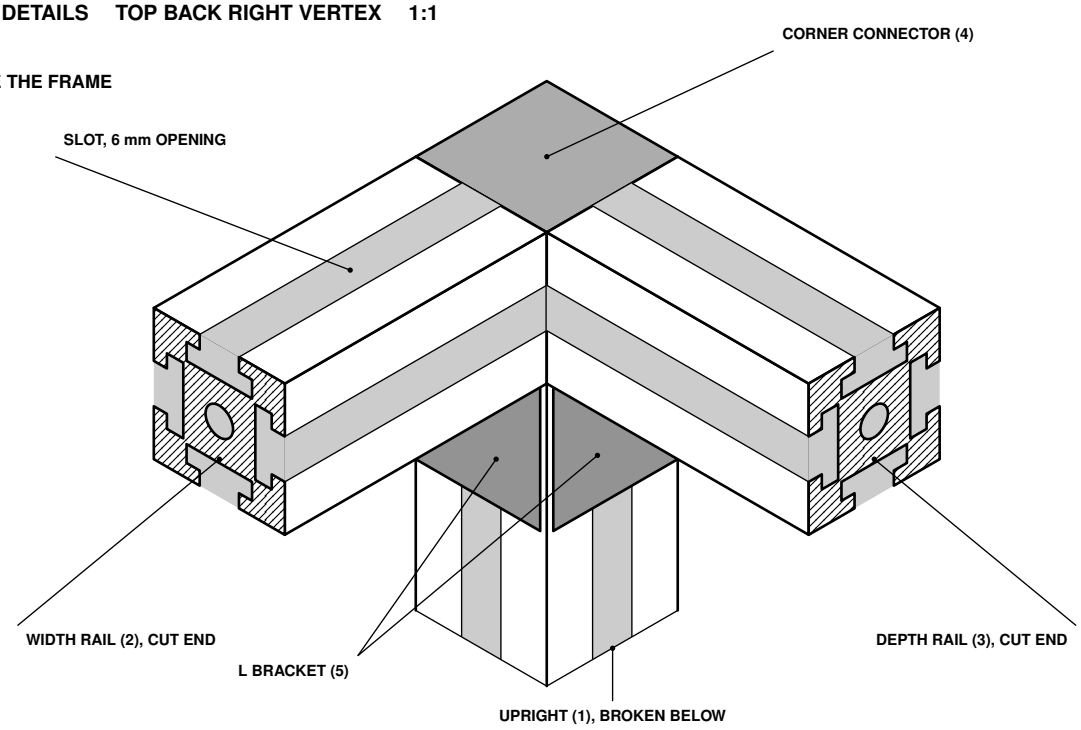
STANDARDS

ISO 5457 · 7200 · 128 · 129-1

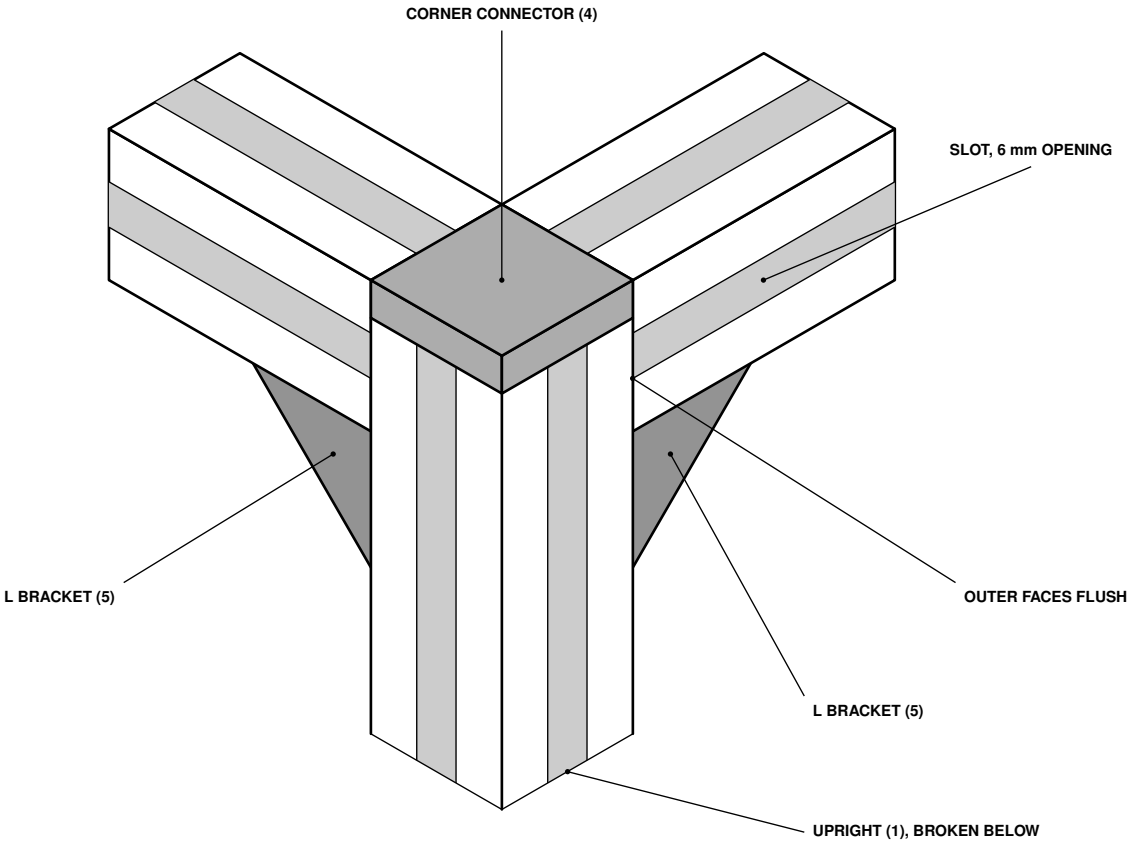
| DRAWING NO. | SCALE | SHEET | DATE | REV. |
|---|---|---|---|---|
| **MR2020-03** | **1:2 / 2:1** | **3 / 5** | 2026-09-13 | **A** |

## CORNER DETAILS  TOP BACK RIGHT VERTEX  1:1

### A  INSIDE THE FRAME




### B  OUTSIDE THE FRAME



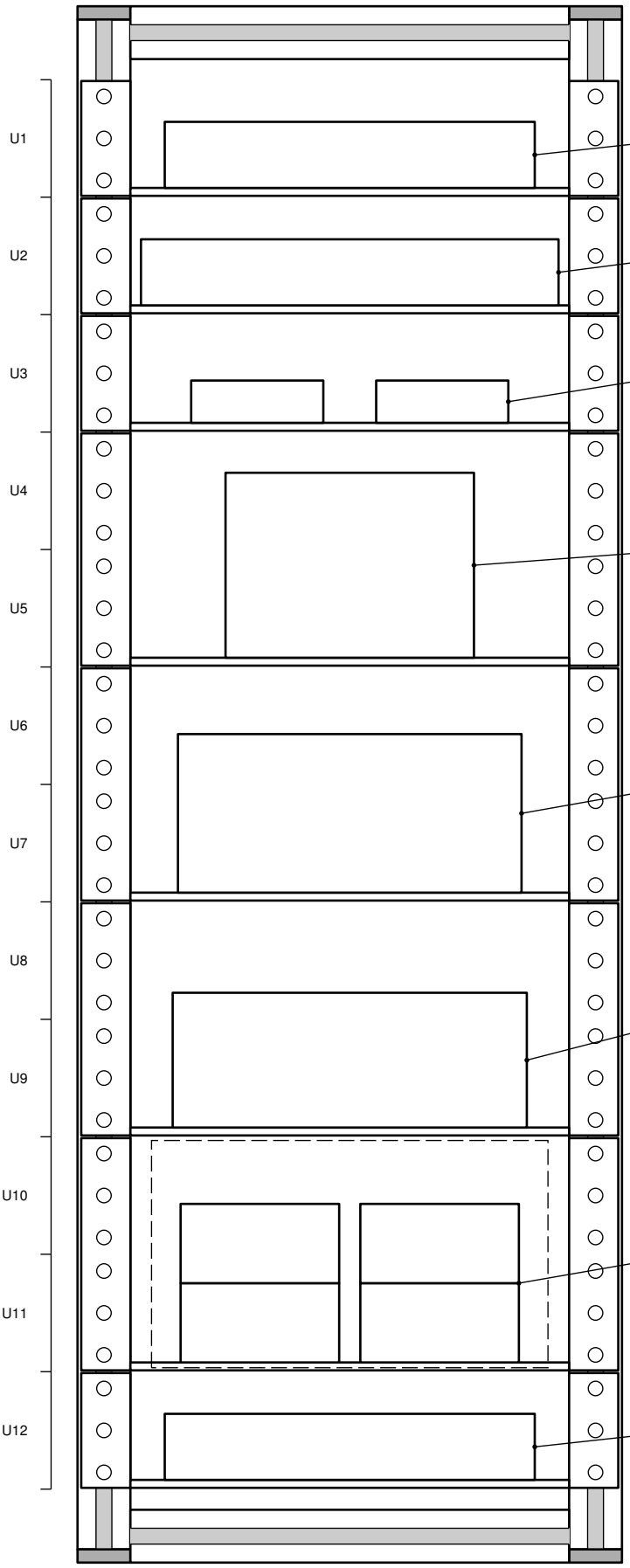

**NOTES**

1. A: seen from inside the frame; the cut ends of the rails show the 2020 profile section.
2. B: seen from outside the frame; the outer faces of the upright and rails are flush.
3. Corner connector and L brackets drawn schematically. Item numbers as on sheets 01 to 03.
4. Each back vertex takes two L brackets under the rails: side plane and back plane.

| | | | | |
|---|---|---|---|---|
| OWNER | | | | |
| **LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION** | | | | |
| TITLE<br>**CORNER DETAILS**<br>Top back right vertex, inside and outside views | | | PROJECTION<br>**ISOMETRIC** | UNIT<br>**mm** |
| MATERIAL<br>Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot | | | STANDARDS<br>ISO 5457 · 7200 · 128 · 129-1 | |
| DRAWING NO.<br>**MR2020-04** | SCALE<br>**1:1** | SHEET<br>**4 / 5** | DATE<br>2026-09-13 | REV.<br>**A** |

## RACK UNIT LAYOUT

FRONT VIEW 1:2.5




**U1 EXHAUST**
2× fan, 140 × 140 × 25

**U2 SWITCH**
TP-Link TL-SG108, 158 × 101 × 25

**U3 SBC PAIR**
2× Orange Pi Zero 3, 50 × 55 × 16

**U4-5 NAS**
SBC + Penta SATA HAT + 4× SSD, 94 × 175 × 70

**U6-7 MINI PC**
AOOSTAR GEM12, 130 × 130 × 60

**U8-9 GPU**
Lying card via OCuLink, ≤250 × 134 × 51

**U10-11 POWER**
ATX PS/2 at the rear (dashed) + 4 bricks, est.

**U12 INTAKE**
2× fan, 140 × 140 × 25

OWNER

**LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION**

| TITLE | PROJECTION | UNIT |
|---|---|---|
| **RACK UNIT LAYOUT**<br>Front view, tray and equipment allocation per U | | **mm** |

| MATERIAL | STANDARDS |
|---|---|
| Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot | ISO 5457 · 7200 · 128 · 129-1 |

| DRAWING NO. | SCALE | SHEET | DATE | REV. |
|---|---|---|---|---|
| **MR2020-05** | **1:2.5** | **5 / 5** | 2026-09-13 | **A** |